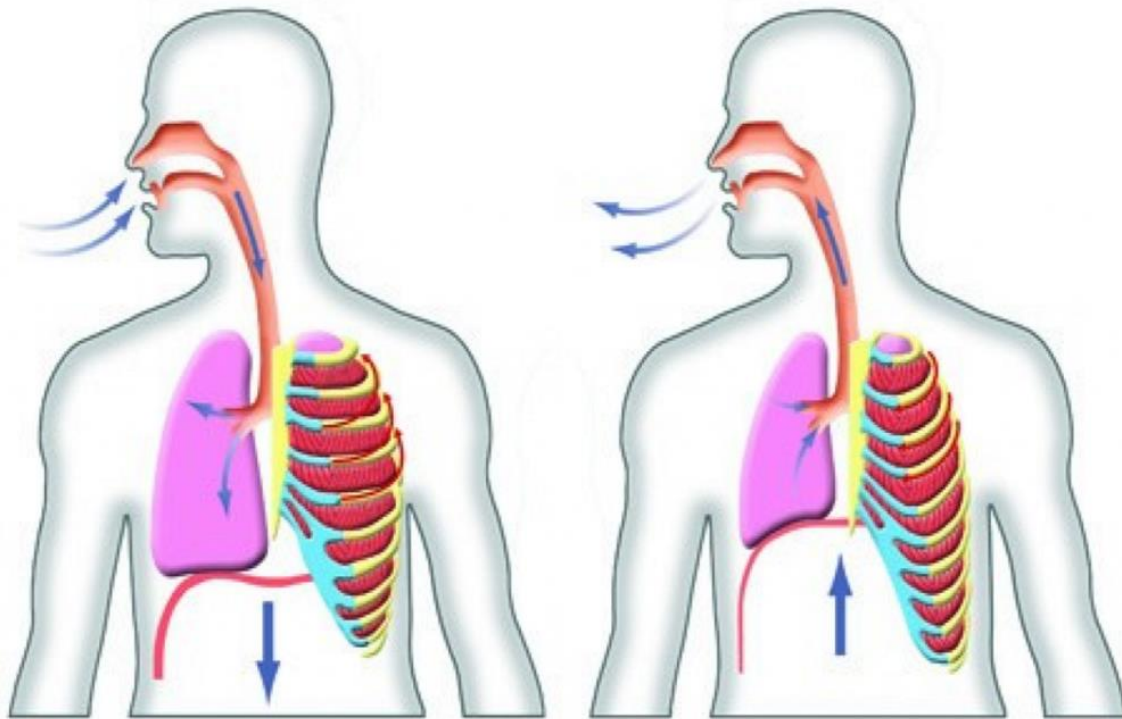


The Structure and Function of the Respiratory System



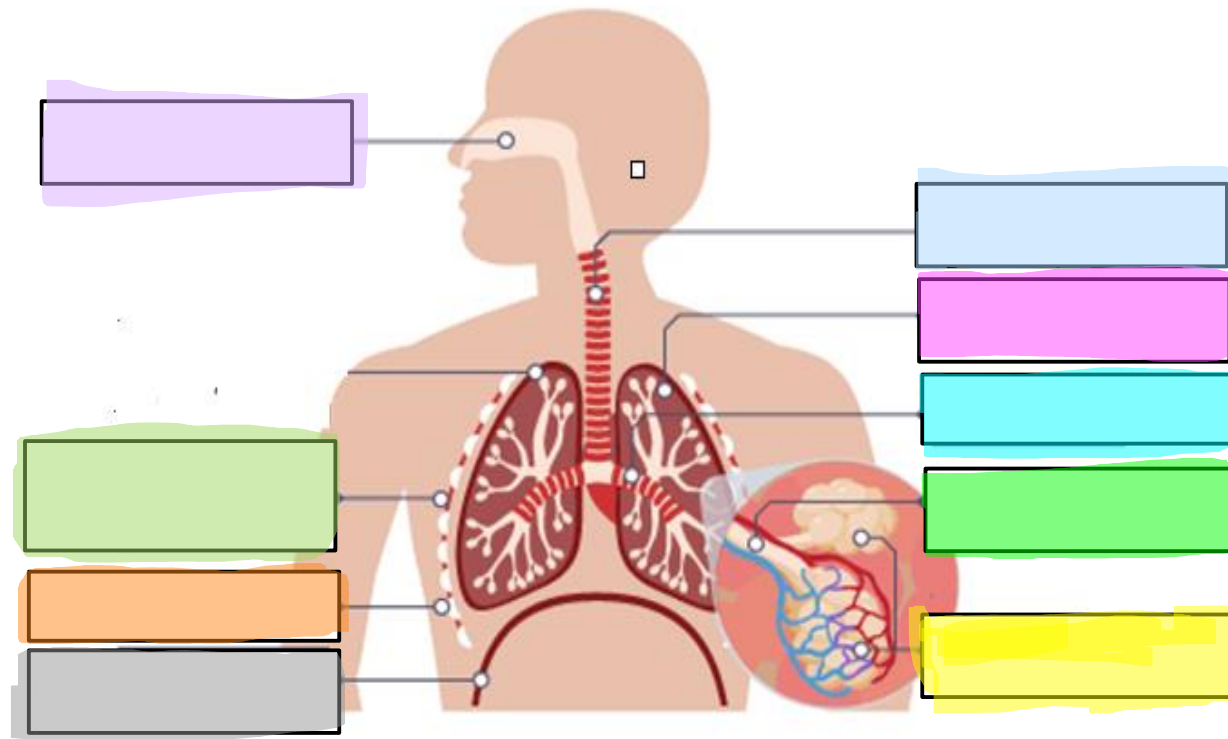
Learning objectives and outcomes

- ▶ To develop our understanding of the structure and function of the respiratory system through **concentration**
- ▶ **5: I know what the function of the respiration system is and some of the parts of its structure.**
- ▶ **6-7: All the above and can explain the pathway of air through the respiratory system. I can also describe what happens during respiration.**
- ▶ **8-9: all the above, and I can explain what the effect of exercise is on the respiratory system. I can also define all key terms.**

The structure of the respiration system

- ▶ The respiratory and cardiovascular system work closely together to maintain a supply of oxygen to the working muscles
- ▶ External Respiratory System=Exchange of gases between lungs and the blood
- ▶ Internal Respiratory System=Exchange of gases between the blood and the cells

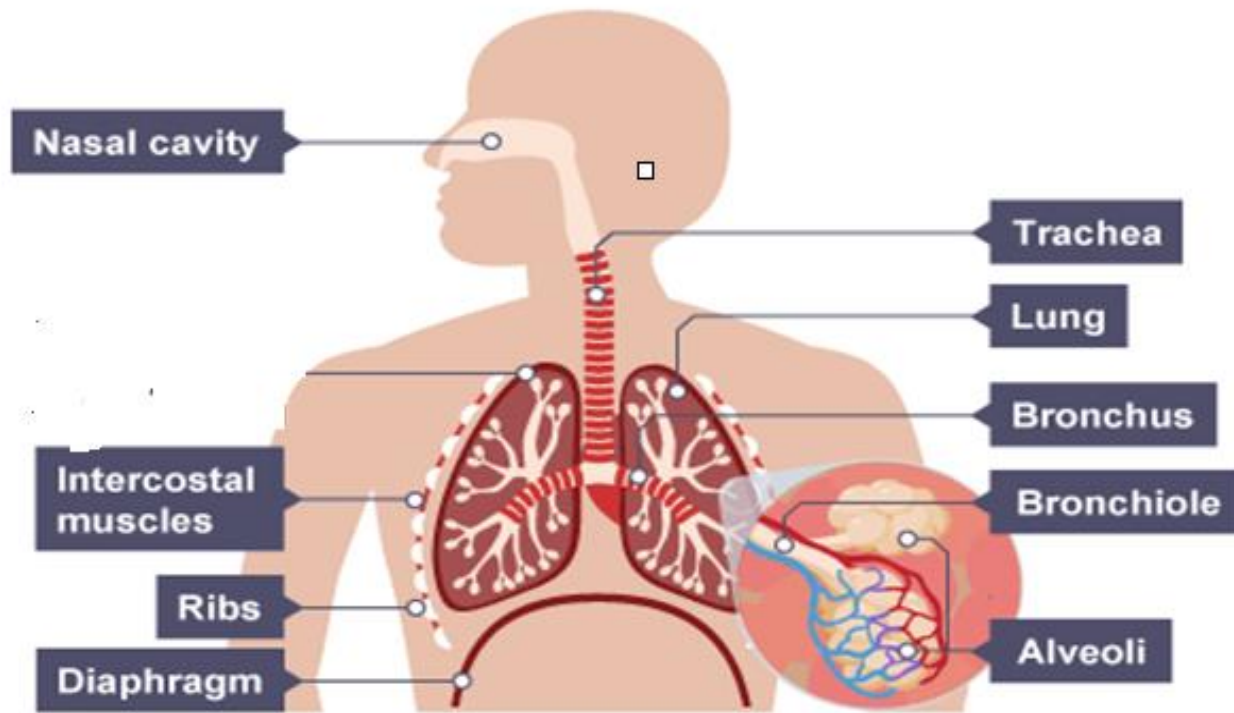
Task One:



Label your diagram using the following labels.

Alveoli, Lung, Bronchus, Bronchiole, Trachea, Ribs, Diaphragm, Nasal Cavity and Intercostal Muscles

Green Pens Out-Check your diagram labels match the image below...



Extension Task: Using the diagram labelled, attempt to describe the pathway of air through the respiratory system

The pathway of air through the respiratory system

- 1) Air enters the body through the nose and mouth and is warmed by the mucus membranes or damp walls.
- 2) The air then enters the trachea/windpipe
- 3) The trachea divides into two bronchi. The right bronchus goes into the right lung, the left bronchus goes into the left lung.
- 4) Each bronchus branches out into smaller tubes called bronchioles. Air travels through these bronchioles.
- 5) At the end of the bronchioles, the air enters one of the many millions of alveoli where gaseous exchange takes place

Extension Question:

1) Describe the characteristics of the alveoli

2) What is the function of the pharynx?

1) Tiny air filled sacs, Millions in the lungs, Enormous surface area (Estimated size of tennis court), Extremely thin walls and are lined by a thin film of water allowing oxygen to dissolve

2) The function of the pharynx is that it is the passage to the digestive system and to the larynx

Simplified Pathway of Air Through the Respiratory System

Mouth/
Nose

Trachea

Bronchi

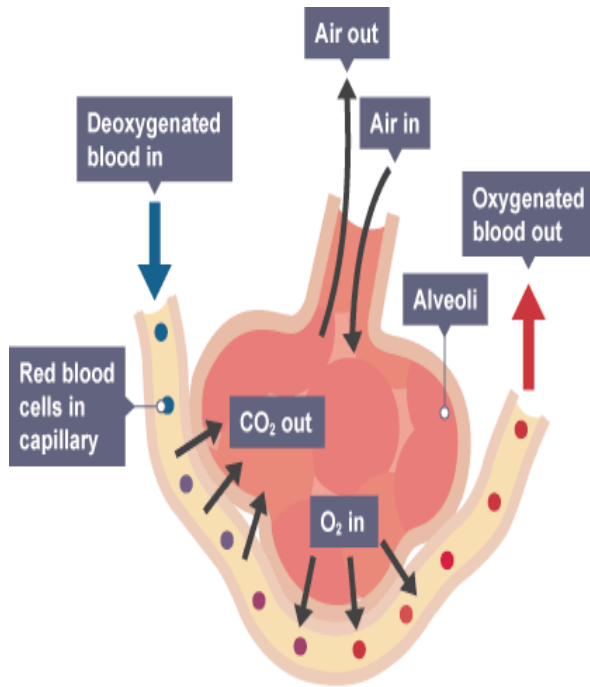
Bronchioles

Alveoli

Red Blood
Cells



Looking at the image below identify and describe what it is



Deoxygenated blood (blood cells blue for purposes of diagram only)

Extension Task: 1) In what vessel does deoxygenated blood travel in?
2) Why is diffusion possible in the lungs?

1) The capillaries

2) Because both the capillaries and alveoli walls are very thin-just one cell thick

Gaseous Exchange

- ▶ Within the alveoli an exchange of gases takes place between the gases inside the alveoli and the blood.

- ▶ Blood arriving in the alveoli:

Higher carbon dioxide concentration produced by the body's cells

Lower oxygen concentration

- ▶ Air in the alveoli:

Lower concentration of carbon dioxide

Higher oxygen concentration

- ▶ Diffusion:

Oxygen moves into the blood by diffusion as this is the movement of gas from an area of high concentration to an area of low concentration (combines with haemoglobin in red blood cells to form oxyhaemoglobin).

Extension Question:

1) Why is the body's oxygen concentration lower when blood arrives in the alveoli?

2) What is the role of haemoglobin?

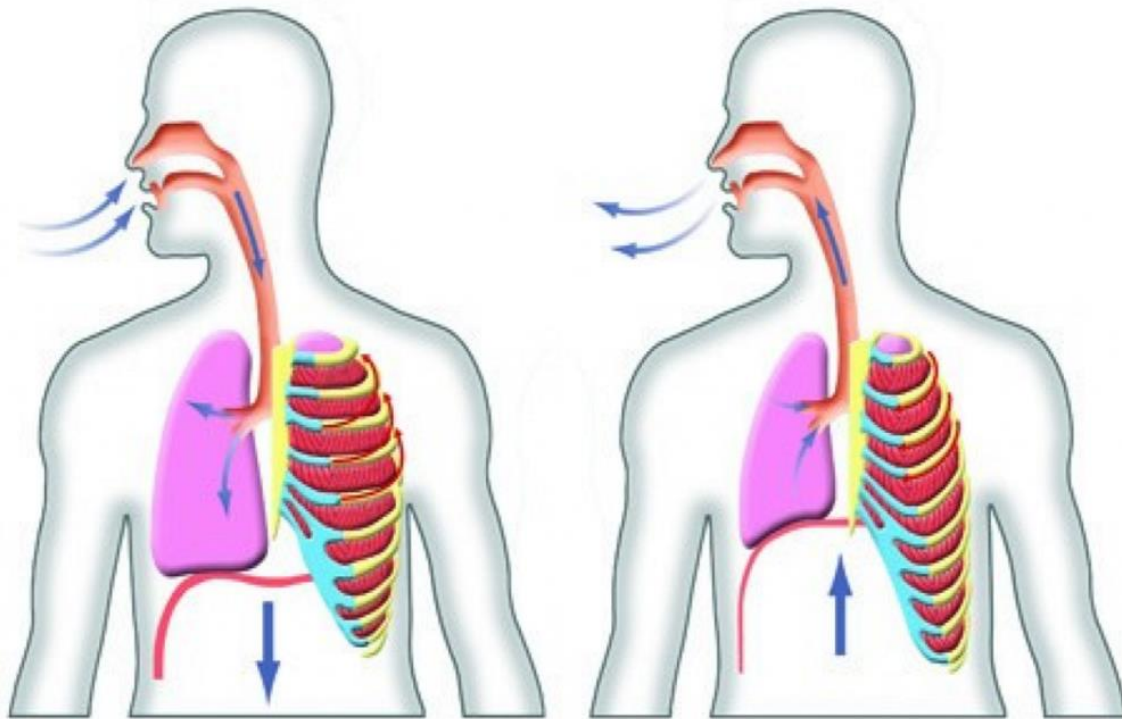
1) Because it has been used for respiration by the body's cells

2) Haemoglobin carries oxygen to be exchanged at the working muscle and carbon dioxide to be exchanged at the lung

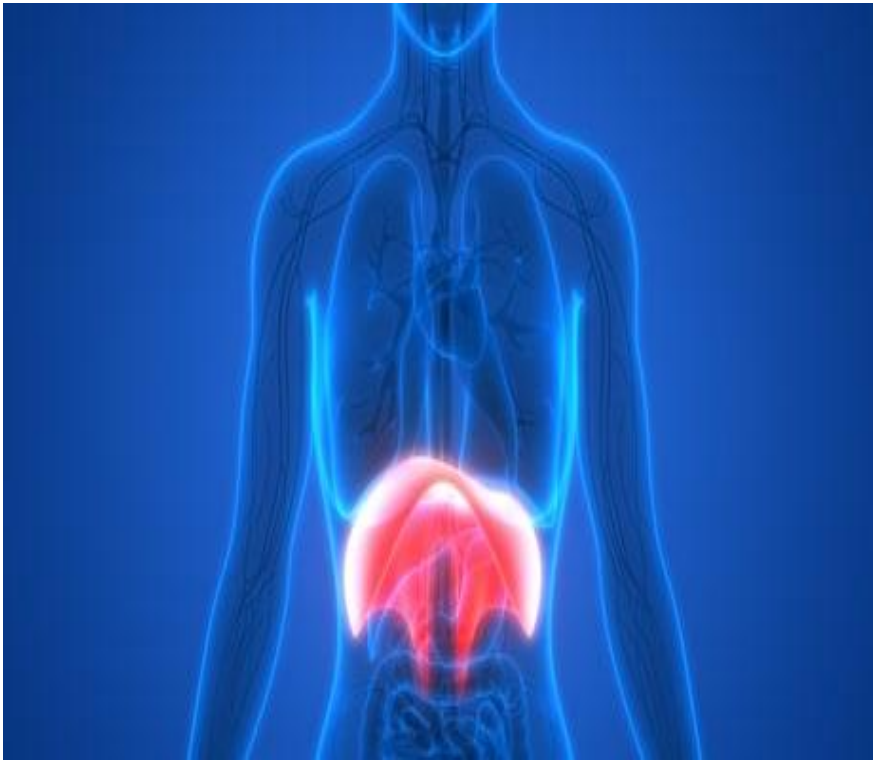
Learning objectives and outcomes

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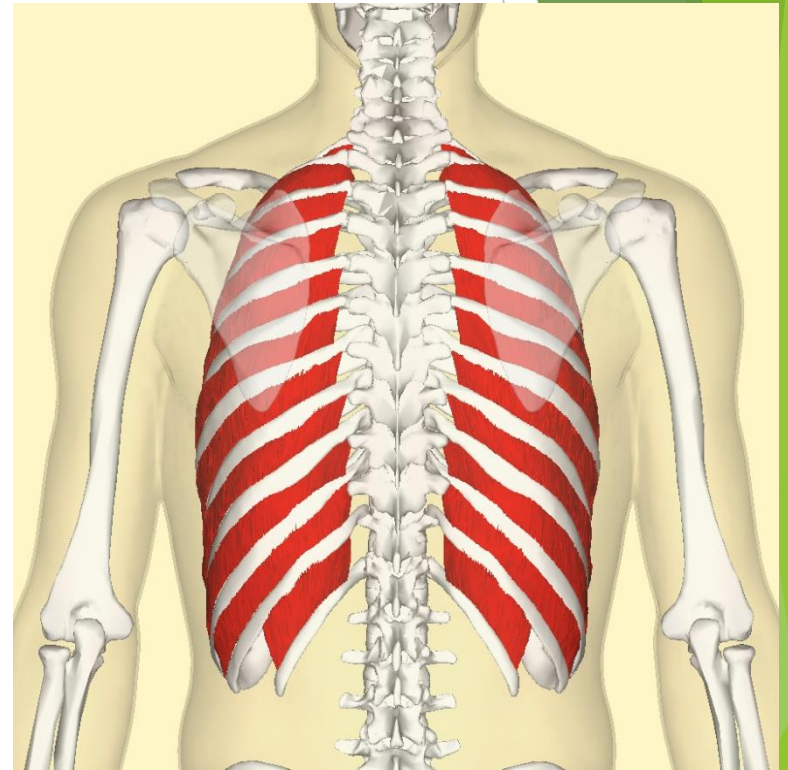
The Structure and Function of the Respiratory System



Can you identify what the
images are of below:



Diaphragm



Intercostal Muscles
(Attached to the ribs)

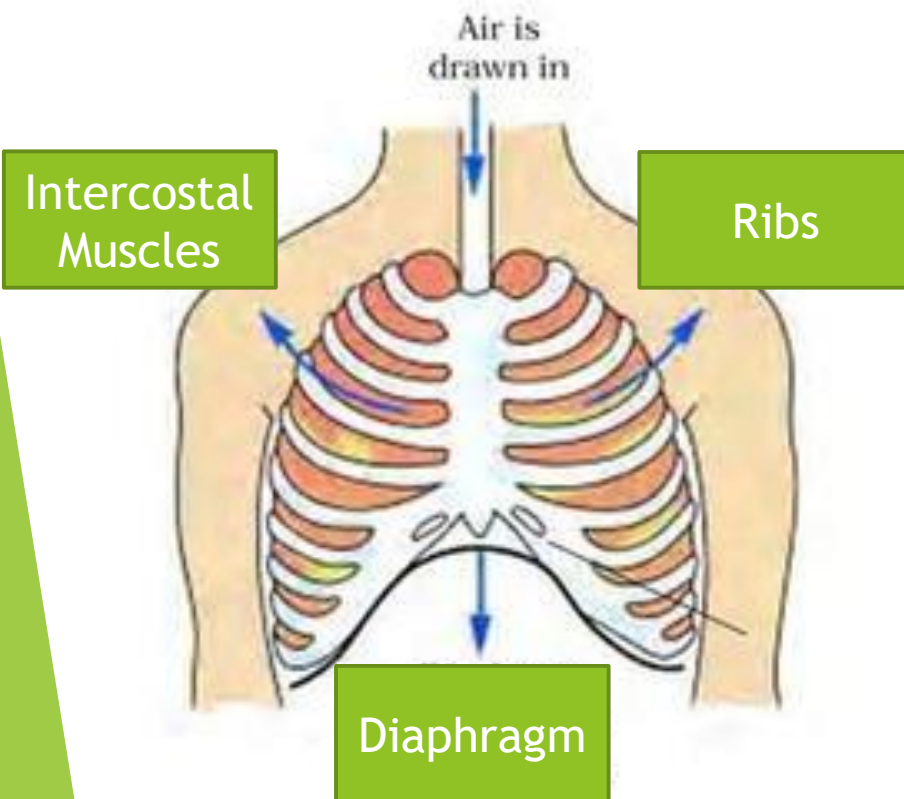
The Role of Respiratory Muscles:

- ▶ Two important structures for breathing:
- ▶ Diaphragm
- ▶ Intercostal muscles

- ▶ Diaphragm-Sheet of muscle that separates the chest cavity from the rest of the body
- ▶ Intercostal muscles-Found between the ribs and they control rib movement

During Inspiration (Breathing In):

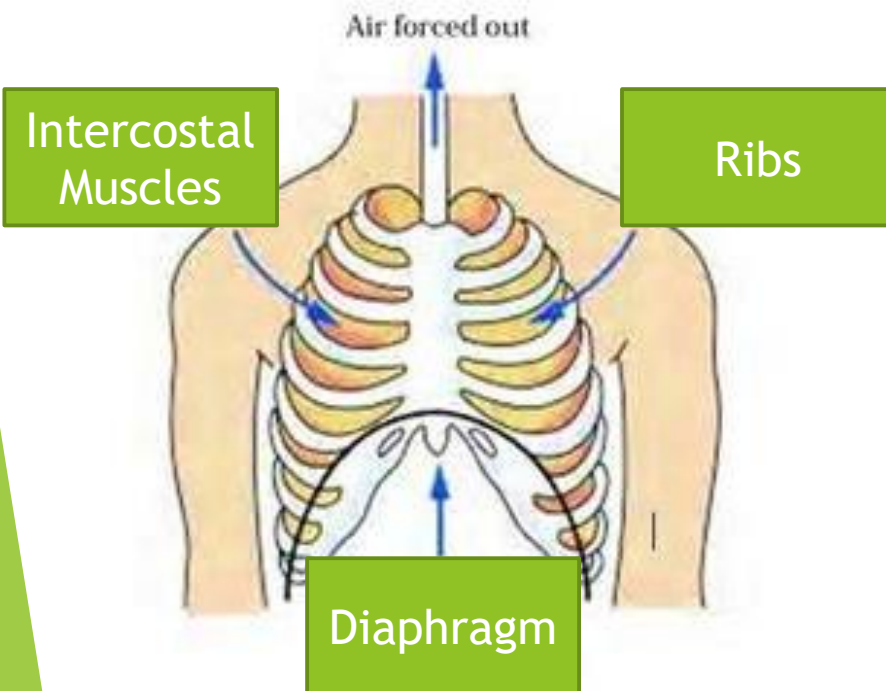
Looking at the image below-can you describe what happens to the diaphragm, chest, ribs, intercostal muscles and lungs during this phase:



- Diaphragm contracts and moves down
- Intercostal muscles contract=Moves the ribs upwards and outwards
- Size of the chest increases
- Space in the lungs increases
- Pressure in the lungs decreases and becomes less than the pressure outside the body
- Gases move from areas of high pressure to areas of low pressure=Air inspired into the lungs

During Expiration (Breathing Out):

Looking at the image below-can you describe what happens to the diaphragm, chest, ribs, intercostal muscles and lungs during this phase
(HINT-LOOK AT THE DESCRIPTION OF BREATHING IN FOR HELP):

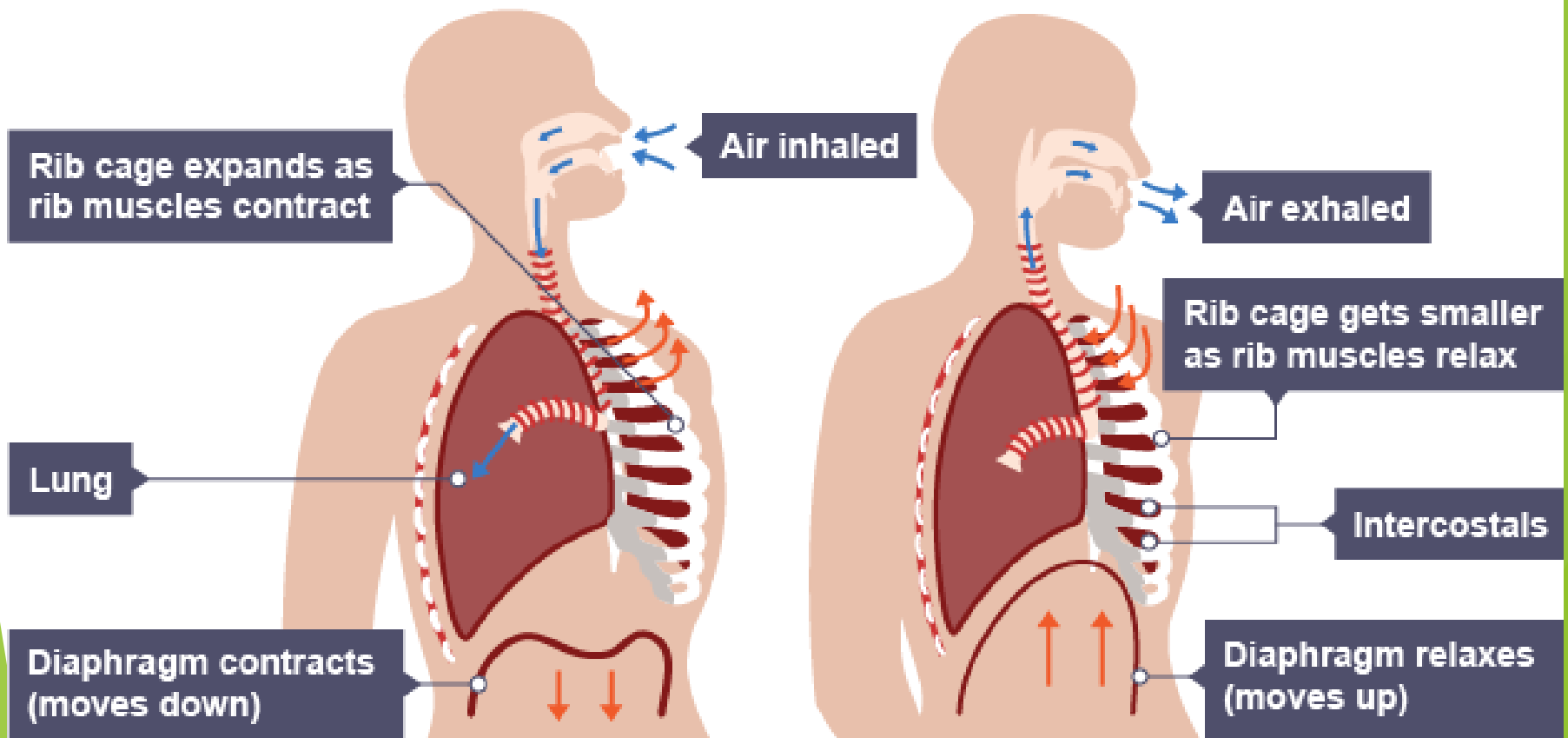


- Diaphragm relaxes
- Intercostal muscles relax=Ribs are lowered
- Size of chest decreases
- Space in the lungs decreases
- Pressure in lungs increases and becomes greater than the pressure outside the body
- Air is forced out to equalise this pressure=Expiration occurs

Simplified...

Inhalation

Exhalation



Identify what the following key terms mean:

► Tidal Volume:

The v_lu_e of a_r that is i_spi_ed or e_pi_ed per br_at_.

The volume of air that is inspired or expired per breath.

► Minute Ventilation:

The volume of air that is inspired or expired in o_e m_n_te

The volume of air that is inspired or expired in one minute

Calculation:

Minute ventilation (VE)= Tidal Volume (TV) X Number of breaths per minute (F)

► Breathing Rate:

The fr_qu_ _cy of br_ _th_n_ measured in br_a_ _s per mi_u_ _

The frequency of breathing measured in breaths per minute.

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